

WRIGHT-PATTERSON AFB, OH, USA

WMO: 745700

Lat: 39.833N

Lon: 84.050W

Elev: 823

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13840

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.5	(c) 8.6	(d) -4.5	(e) 4.5	(f) 4.4	(g) 0.8	(h) 5.9	(i) 10.2	(j) 26.6	(k) 34.3	(l) 24.4	(m) 32.9	(n) 6.9	(o) 260	(p) 0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.4	(c) 90.8	(d) 74.3	(e) 88.5	(f) 73.2	(g) 86.2	(h) 71.9	(i) 77.2	(j) 86.2	(k) 75.8	(l) 84.4	(m) 74.5	(n) 82.5	(o) 9.5	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	134.4	81.4	73.2	127.0	79.9	72.1	122.5	78.9	41.1	85.8	39.7	84.1	38.5	82.7	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.7	(b) 19.6	(c) 17.5	(e) -5.4	(f) 94.2	(g) 7.7	(h) 3.3	(i) -10.9	(j) 96.5	(k) -15.4	(l) 98.5	(m) -19.7	(n) 100.3	(o) -25.2	(p) 102.7		
(4) 22.7	(5) 19.6	(6) 17.5	(7) -5.8	(8) 79.7	(9) 7.5	(10) 1.6	(11) -11.2	(12) 80.8	(13) -15.6	(14) 81.8	(15) -19.8	(16) 82.6	(17) -25.3	(18) 83.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	28.6	33.1	41.0	52.6	63.2	71.2	74.5	73.0	66.4	54.9	43.1	34.2		
	DBStd	18.43	12.79	11.54	11.14	9.46	8.36	6.27	5.43	4.99	7.26	8.71	9.46	10.59		
	HDD50	2366	673	485	327	80	6	0	0	0	0	47	248	500		
	HDD65	5327	1130	894	747	382	137	19	2	4	70	331	656	955		
	CDD50	3494	8	11	47	159	416	635	761	714	492	199	42	10		
	CDD65	980	0	0	2	11	83	204	297	253	112	18	0	0		
	CDH74	8619	0	1	19	115	742	1740	2694	2135	1002	166	5	0		
	CDH80	2828	0	0	2	11	180	588	1003	690	323	31	0	0		
(14) Wind	WSAvg	7.7	9.6	9.3	9.3	9.3	7.4	6.6	6.0	5.6	5.7	7.1	8.0	8.4		
(15) Precipitation	PrecAvg	41.6	3.1	2.1	3.1	4.2	4.6	4.3	4.6	3.2	3.3	3.1	2.7	3.3		
	PrecMax	60.8	9.3	4.6	6.9	9.6	8.1	7.8	8.0	6.0	9.9	5.9	5.9	8.5		
	PrecMin	30.2	1.1	0.7	0.9	1.2	1.6	1.7	1.9	0.7	0.8	0.8	0.9	1.0		
	PrecStd	6.9	1.7	1.1	1.2	1.8	1.9	1.5	1.5	1.3	2.1	1.5	1.3	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.4	69.8	76.9	81.4	87.9	92.7	95.9	91.7	92.1	84.1	72.6	64.4		
		MCWB	58.1	59.1	61.4	64.8	71.5	73.8	77.0	74.4	72.2	68.0	61.8	59.0		
	2%	DB	58.9	61.3	71.3	77.5	84.5	89.1	91.1	89.6	87.5	78.6	67.1	59.3		
		MCWB	55.6	54.2	58.6	62.1	68.7	72.7	75.7	73.9	70.7	64.8	58.5	54.5		
	5%	DB	53.7	55.6	65.4	73.2	81.8	86.3	88.6	86.5	83.9	74.5	63.2	54.8		
		MCWB	51.3	50.1	55.6	59.9	67.8	71.4	74.1	72.6	69.4	62.8	56.1	51.2		
	10%	DB	47.0	50.3	59.8	69.5	78.7	83.6	85.9	84.0	80.3	70.3	58.8	50.2		
		MCWB	43.2	45.0	52.4	58.3	66.4	70.0	72.9	71.3	66.9	61.3	52.7	46.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.6	61.0	64.0	68.2	73.9	76.8	80.1	78.2	76.1	71.9	64.2	60.5		
		MCDB	61.7	67.5	72.8	78.7	83.7	86.4	90.2	86.4	86.5	80.7	68.8	63.8		
	2%	WB	56.0	56.3	60.3	65.2	71.7	75.1	78.0	76.6	73.9	68.5	60.6	55.9		
		MCDB	59.0	59.9	67.8	73.2	80.7	84.9	87.4	84.7	83.2	73.4	65.1	58.5		
	5%	WB	50.9	50.6	57.4	62.5	70.1	73.7	76.4	75.1	72.0	65.6	57.3	51.7		
		MCDB	53.4	55.2	63.1	69.5	78.0	82.6	85.1	82.7	79.3	72.1	61.4	54.7		
	10%	WB	43.5	44.9	53.3	60.0	68.1	72.3	75.0	73.5	70.1	62.4	53.6	46.5		
		MCDB	45.5	49.5	59.1	66.9	75.7	80.2	82.8	80.5	76.1	68.9	58.0	49.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	17.0	19.0	21.6	20.5	19.6	19.4	19.9	21.6	20.9	17.9	14.5		
		MCDBR	18.7	24.6	26.5	26.1	24.1	22.9	21.7	22.1	25.1	26.1	23.8	19.7		
		MCWBR	16.0	18.6	16.2	15.0	11.3	9.6	8.4	9.0	10.4	13.7	16.5	16.5		
	5% WB	MCDBR	17.8	22.6	22.3	21.5	19.8	19.4	19.2	18.4	19.6	21.3	19.1	17.8		
		MCWBR	16.2	19.1	15.6	14.3	10.5	9.4	8.5	8.6	9.6	12.8	15.6	16.4		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.327	0.348	0.393	0.439	0.448	0.450	0.445	0.395	0.357	0.325	0.304		
	taud		2.440	2.342	2.365	2.254	2.166	2.193	2.193	2.189	2.309	2.410	2.457	2.455		
	Ebn at Noon		274	281	287	279	268	265	263	261	268	267	263	265		
	Edh at Noon		25	32	35	42	47	46	45	44	37	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		539	814	1130	1522	1743	1926	1883	1733	1415	993	667	456		
	RadStd		49	95	98	126	145	128	110	87	130	91	81	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	+117

Nomenclature: See separate page